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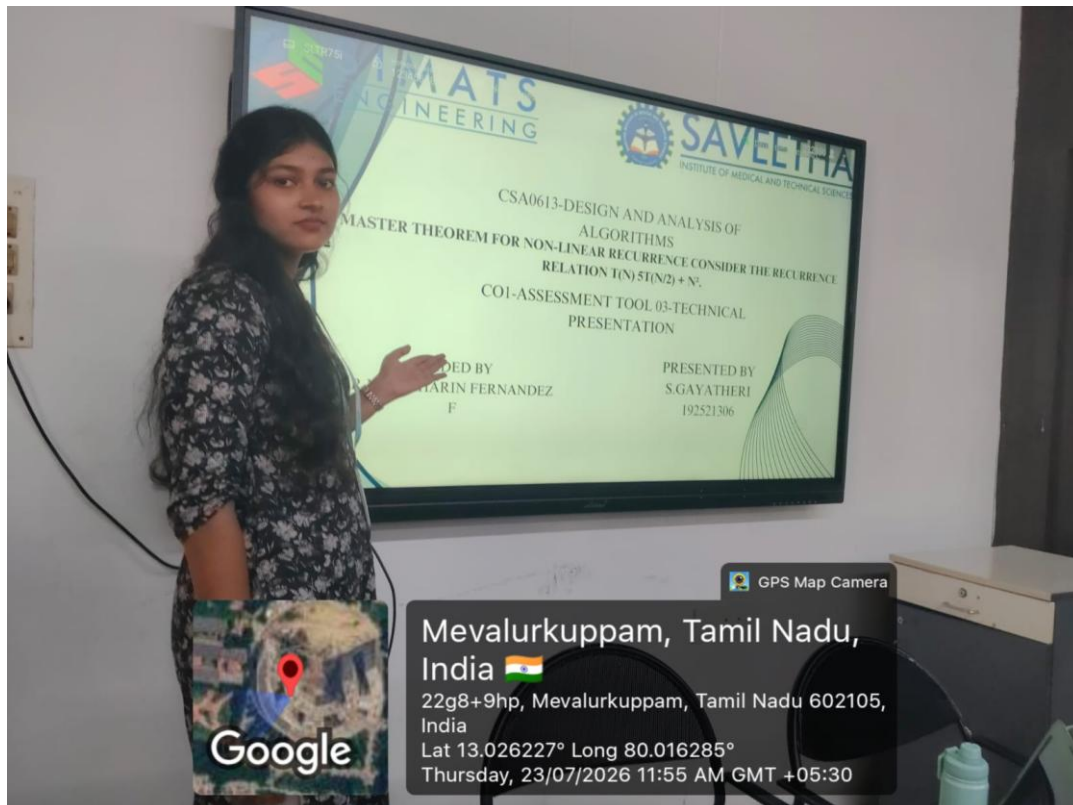
COURSE CODE: CSA0619

REG NO: 192521306

**COURSE NAME: Design and
Analysis of Algorithms**

C01-ASSESSMENT TOOL 03 -TECHNICAL PRESENTATION

**TITLE: "Master Theorem for Non-Linear Recurrence Consider the recurrence relation
 $T(n) = 5T(n/2) + n^2$ "**



PROJECT DESCRIPTION

This project focuses on analysing the Master Theorem for Non-Linear Recurrence Relations using the recurrence relation $T(n) = 5T(n/2) + n^2$. The project demonstrates how the Master Theorem can be applied to determine the time complexity of recursive algorithms. By comparing the recursive term $5T(n/2)$ with the non-recursive work n^2 , the project identifies the appropriate case of the Master Theorem and derives the asymptotic complexity. It also explains how recurrence relations help in analyzing the efficiency and performance of divide-and-conquer algorithms.